

ROBIEI, Switzerland

WMO: 067510

Lat: 46.4431N

Lon: 8.5133E

Elev: 6204

StdP: 11.69

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.1	11.1	-19.7	2.4	19.6	-13.8	3.3	21.0	25.5	23.2	22.7	24.2	7.4	350	0.672

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	67.2	54.0	64.8	53.2	62.4	52.3	57.2	62.5	55.6	61.1	54.1	59.6	4.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.9	81.0	59.6	53.3	76.2	58.2	51.7	71.9	56.7	27.4	62.7	26.2	61.2	25.2	59.7	63.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.4	16.6	13.8	DB	5.1	71.4	4.5	3.0	1.9	73.6	-0.7	75.4	-3.2	77.0	-6.4	79.2	(4)
(5)			WB	2.6	59.9	3.8	2.1	-0.1	61.4	-2.4	62.6	-4.6	63.7	-7.3	65.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	38.9	25.8	25.4	29.8	35.3	41.4	50.9	54.5	53.6	47.4	41.2	32.8	27.9	(6)
(7)		DBStd	12.24	7.40	8.02	7.18	6.02	5.93	6.15	4.99	5.23	5.64	6.63	7.01	7.42	(7)
(8)		HDD50	4473	749	688	625	441	273	62	18	23	114	280	516	684	(8)
(9)		HDD65	9520	1214	1108	1090	890	731	424	325	355	529	739	966	1149	(9)
(10)		CDD50	430	0	0	0	0	7	89	158	134	35	6	1	0	(10)
(11)		CDD65	2	0	0	0	0	0	1	0	1	0	0	0	0	(11)
(12)		CDH74	3	0	0	0	0	0	2	0	1	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	4.4	5.3	5.2	5.1	4.4	4.8	4.4	3.9	3.6	3.7	3.3	4.2	5.0	(14)
(15)	Precipitation	PrecAvg	71.8	3.4	3.3	3.7	6.4	7.6	7.0	5.8	7.0	7.9	7.6	8.3	3.7	(15)
(16)		PrecMax	99.2	10.3	11.1	11.9	17.8	21.8	16.3	14.1	12.4	24.1	27.3	29.7	9.2	(16)
(17)		PrecMin	39.8	0.7	0.4	0.3	0.8	1.7	1.9	1.9	1.9	0.5	0.4	0.7	0.2	(17)
(18)		PrecStd	14.9	2.3	2.4	2.6	4.5	4.6	3.5	2.6	2.4	6.1	5.7	7.2	2.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.1	50.2	47.6	52.9	60.1	70.2	69.7	71.1	64.1	58.9	54.1	46.2	(19)
MCWB			33.4	34.8	34.1	40.3	48.1	53.2	56.0	53.8	50.0	43.8	39.3	33.0	(20)	
2%		DB	43.0	44.6	44.9	48.8	56.0	66.2	66.8	66.9	60.2	55.4	48.8	43.4	(21)	
		MCWB	32.1	32.8	33.5	38.7	44.8	52.5	54.0	54.4	49.7	42.6	37.6	32.2	(22)	
5%		DB	40.0	40.3	42.6	46.1	52.7	63.3	64.4	64.1	57.8	52.4	45.3	40.6	(23)	
		MCWB	30.9	30.5	32.5	37.0	43.3	51.8	53.6	53.5	48.3	41.5	35.7	30.7	(24)	
10%		DB	36.7	36.8	40.1	43.9	49.9	60.4	62.1	61.3	55.4	49.7	42.1	38.0	(25)	
		MCWB	28.6	28.6	31.6	36.1	41.7	50.5	52.5	52.4	47.5	41.0	33.9	29.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.3	36.6	36.9	42.4	50.1	57.4	60.0	59.4	54.5	48.5	42.3	36.2	(27)
MCD B			42.5	45.4	43.3	48.6	57.6	63.4	65.3	65.3	58.6	52.9	48.0	42.4	(28)	
2%		WB	33.6	33.6	35.2	40.2	46.8	55.4	57.4	57.1	52.4	46.5	39.3	34.0	(29)	
		MCD B	40.8	42.7	40.9	45.7	53.1	61.8	62.7	62.4	56.8	49.0	46.3	39.3	(30)	
5%		WB	31.4	31.2	34.0	38.7	44.7	53.3	55.7	55.4	50.8	45.2	37.3	32.3	(31)	
		MCD B	38.0	38.9	39.5	44.0	50.4	61.0	61.3	60.6	55.0	48.1	42.0	37.1	(32)	
10%		WB	29.2	29.1	32.6	37.2	43.1	51.5	54.2	53.9	49.1	43.7	35.8	30.7	(33)	
		MCD B	35.0	35.1	38.1	42.5	48.2	58.7	59.8	58.7	53.4	47.0	39.8	35.9	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.1	9.4	9.2	8.9	8.9	10.1	10.3	9.3	9.1	8.6	7.7	8.6	(35)
MCDBR			12.1	12.6	11.4	11.1	12.6	13.6	13.0	12.4	11.7	12.3	11.4	11.2	(36)	
MCWBR			8.2	7.5	7.1	6.8	7.6	8.6	8.5	8.4	8.2	7.4	6.6	7.2	(37)	
5% WB		MCDBR	11.3	11.0	9.3	9.4	10.4	12.0	11.6	10.3	10.1	8.6	8.5	9.4	(38)	
		MCWBR	8.5	7.7	6.4	6.6	7.3	8.8	8.6	8.0	7.9	6.6	6.6	7.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.232	0.257	0.299	0.344	0.361	0.370	0.366	0.356	0.333	0.307	0.265	0.231	(40)	
taud		2.333	2.279	2.222	2.204	2.273	2.295	2.303	2.345	2.401	2.453	2.447	2.369	(41)		
Ebn at Noon		282	294	294	288	286	284	283	282	279	272	267	270	(42)		
Edh at Noon		23	30	37	42	41	40	40	36	31	25	21	20	(43)		
(44)	All-Sky Solar Radiation	RadAvg	509	816	1202	1418	1527	1795	1797	1556	1211	800	496	425	(44)	
(45)		RadStd	64	96	173	227	151	156	153	135	117	96	75	55	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+1.26	N/A	N/A	N/A	N/A	N/A
(47) Regional (121 neighbors)		N/A	N/A	N/A	+1.31	+0.76	+0.58	N/A	N/A	+103	+27

Nomenclature: See separate page